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Kinematic structures for robotic microsurgical procedures

Abstract

Apparatus and methods are described for performing a procedure using a robotic unit. A tool-actuation arm is driven to move linearly, to thereby move at least the portion of the tool linearly with respect to the end effector. The tool-actuation arm is driven to become retracted to a given distance from the tool mount, thereby causing the tool-actuation arm to fold automatically by actuating an automatic tool-actuation arm folding mechanism. Other applications are also described.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of PCT application no. PCT/IB2022/055086 to Gil et al. filed May 31, 2022, entitled "Kinematic structures and sterile drapes for robotic microsurgical procedures" (published as WO 22/254335), which claims priority from: U.S. Provisional Patent Application No. 63/195,429 to Gil et al., filed Jun. 1, 2021, entitled "Kinematic structures for robotic microsurgical procedures," and U.S. Provisional Patent Application No. 63/229,593 to Gil et al., filed Aug. 5, 2021, entitled "Sterile drapes for robotic microsurgical procedures." Both of the above-referenced U.S. Provisional applications are incorporated herein by reference.

FIELD OF EMBODIMENTS OF THE INVENTION
(1) Some applications of the present invention generally relate to medical apparatus and methods. Specifically, some applications of the present invention relate to apparatus and methods for performing microsurgical procedures in a robotic manner.
BACKGROUND
(2) Cataract surgery involves the removal of the natural lens of the eye that has developed an opacification (known as a cataract), and its replacement with an intraocular lens. Such surgery typically involves a number of standard steps, which are performed sequentially.
(3) In an initial step, the patient's face around the eye is disinfected (typically, with iodine solution), and their face is covered by a sterile drape, such that only the eye is exposed. When the disinfection and draping has been completed, the eye is anesthetized, typically using a local anesthetic, which is administered in the form of liquid eye drops. The eyeball is then exposed, using an eyelid speculum that holds the upper and lower eyelids open. One or more incisions (and typically two or three

incisions) are made in the cornea of the eye. The incision(s) are typically made using a specialized blade, which is called a keratome blade. At this stage, lidocaine is typically injected into the anterior chamber of the eye, in order to further anesthetize the eye. Following this step, a viscoelastic injection is applied via the corneal incision(s). The viscoelastic injection is performed in order to stabilize the anterior chamber and to help maintain eye pressure during the remainder of the procedure, and also in order to distend the lens capsule.

(4) In a subsequent stage, known as capsulorhexis, a part of the anterior lens capsule is removed. Various enhanced techniques have been developed for performing capsulorhexis, such as laser-assisted capsulorhexis, zepto-rhexis (which utilizes precision nano-pulse technology), and marker-assisted capsulorhexis (in which the cornea is marked using a predefined marker, in order to indicate the desired size for the capsule opening).

(5) Subsequently, it is common for a fluid wave to be injected via the corneal incision, in order to dissect the cataract's outer cortical layer, in a step known as hydrodissection. In a subsequent step, known as hydrodelineation, the outer softer epi-nucleus of the lens is separated from the inner firmer endo-nucleus by the injection of a fluid wave. In the next step, ultrasonic emulsification of the lens is performed, in a process known as phacoemulsification. The nucleus of the lens is broken initially using a chopper, following which the outer fragments of the lens are broken and removed, typically using an ultrasonic phacoemulsification probe. Further typically, a separate tool is used to perform suction during the phacoemulsification. When the phacoemulsification is complete, the remaining lens cortex (i.e., the outer layer of the lens) material is aspirated from the capsule. During the phacoemulsification and the aspiration, aspirated fluids are typically replaced with irrigation of a balanced salt solution, in order to maintain fluid pressure in the anterior chamber. In some cases, if deemed to be necessary, then the capsule is polished. Subsequently, the intraocular lens (IOL) is inserted into the capsule. The IOL is typically foldable and is inserted in a folded configuration, before unfolding inside the capsule. At this stage, the viscoelastic is removed, typically using the suction device that was previously used to aspirate fluids from the capsule. If necessary, the incision(s) is sealed by elevating the pressure inside the bulbus oculi (i.e., the globe of the eye), causing the internal tissue to be pressed against the external tissue of the incision, such as to force closed the incision.

SUMMARY

(6) In accordance with some applications of the present invention, a robotic system is configured for use in a microsurgical procedure, such as intraocular surgery. Typically, when used for intraocular surgery, the robotic system includes first and second robotic units. For some applications, each of the robotic units includes an end effector, which is typically configured to securely hold any one of a plurality of different tools thereupon. For some applications, the end effector is coupled to a tool mount, which is configured to hold the tools (directly or indirectly). Typically, the end effector is configured to insert the tool into the patient's eye such that entry of the tool into the patient's eye is via an incision point, and the tip of the tool is disposed within the patient's eye.

(7) For some applications, two multi-jointed arms (i.e., arms containing a plurality of links which are connected to each other via joints) are disposed on a single side of end effector and are configured to moveably support the end effector. Typically, a plurality of arm-motors are associated with the two multi-jointed arms. For some applications, the robotic unit is configured to rotate a tool about its own axis, such as to compensate for rolling of the end effector with respect to the base of the robotic unit. It is typically desirable to prevent a tool (and in particular a tool that is not rotationally symmetrical) from rolling with respect to a patient's eye. For some applications, rather than preventing the end effector from rotating with respect to the base of the robotic unit, the end effector is permitted to roll with respect to the base, but such rolling of the end effector is compensated for by rolling the tool about its own axis with respect to the end effector. For some applications, the robotic unit is configured to rotate a tool about its own axis, for an alternative or

additional reason, e.g., in order to perform a surgical maneuver.

(8) Typically, the robotic unit is actively driven to move the end effector along the x-, y-, and z-axes, as well as through pitch and yaw angular movements, whereas rolling of the end effector is an unwanted by-product of such movements. For some applications, a computer processor calculates the amount of roll that the tool should undergo with respect to the end effector. For example, the computer processor may calculate that owing to movement of the multi-jointed arms (e.g., translational movement along the x-, y-, and/or z-axes, and/or pitch and/or yaw angular movement), the end effector will undergo roll of +20 degrees with respect to base. In response thereto, the computer processor may drive the tool to rotate about its own axis with respect to end effector by −20 degrees.

(9) For some applications, as an alternative to or in addition to rolling the tool with respect to the end effector, the end effector itself is rolled. Typically, in such cases, the end effector is rolled about an axis that is not coaxial with the longitudinal axis of the tool. Thus, the end effector undergoes rolling about an axis that is eccentric with respect to its longitudinal axis. For some applications, the robotic unit includes an end-effector motor, which is configured to roll the end effector about the eccentric axis. Typically, the robotic unit includes at least five arm motors. For some applications, the computer processor drives the arms to move such as to compensate for the axis about which the end effector is rolled not being coaxial with the tool axis. In this manner, although the end effector is rotated about the eccentric axis, the tool itself is rolled about its own axis.

(10) For some such applications, each of the multi-jointed arms includes a rotatable arched link in a vicinity of end effector. The rotatable arched link is configured to rotate such as to accommodate rolling of the end effector about axis. Typically, as the end effector is rotated, the end effector pushes the arched link causing it to rotate, such that the end effector becomes accommodated by concavely curved surfaces of the arched links. Such accommodation of the rolling of the end effector is typically desirable, particularly in view of the robotic unit being configured such that the rolling of the end effector is eccentric with respect to its own axis. For example, if in place of the rotatable arched link there was a straight link disposed perpendicularly with respect to end effector axis, then the end effector could only be rotated through a relatively small range of angle before being blocked by the link. By contrast, using a configuration as described herein, the end effector is typically able to roll through more than 180 degrees, e.g., more than 250 degrees, or more than 300 degrees, about the eccentric axis.

(11) For some applications, a sterile drape is provided between (a) the robotic arms and the end effector, which are disposed within a non-sterile zone on a first side of the sterile drape and (b) the tool mount and the tool which are disposed within a sterile zone on a second side of the sterile drape. Typically, the sterile drape is disposed around and sealed with respect to a drape plate. For some applications, the drape plate is couplable to the end effector, and is coupled to (or couplable to) the tool mount. The drape plate typically acts as an interface between (a) the robotic arms and the end effector, which are disposed within a non-sterile zone on a first side of the sterile drape and (b) the tool mount and the tool which are disposed within a sterile zone on a second side of the sterile drape.

(12) For some applications, a tool motor is disposed on the end effector, within the non-sterile zone. The tool motor typically directly drives a motion-transmission portion (such as a pin or a shaft) to move (e.g., to rotate). The motion-transmission portion is configured to transmit motion of the motor to a first gear (e.g., a spur gear (i.e., a gear wheel) or a worm gear) and the first gear drives the tool to rotate with respect to the end effector by driving a second gear (which is typically a spur gear (i.e., a gear wheel) to rotate. (In accordance with respective applications, the second gear is built into the tool itself or can be built into or coupled to the tool sleeve.) Typically, the motion-transmission portion is mechanically coupled to the first gear in such a manner that the interface between the motion-transmission portion and the first gear wheel is sealed (e.g., via an O-ring). Thus, rotational motion of the tool with respect to the end effector is generated by the motor,

which is disposed within the non-sterile zone. The rotational motion that is generated by the motor is transmitted to the tool via an interface that maintains the seal between the non-sterile and sterile zones.

(13) For some applications, a linear tool motor is disposed within the non-sterile zone. The linear tool motor typically drives a tool-actuation arm to move linearly. The tool-actuation arm is typically disposed within the non-sterile zone and is configured to push a portion of a tool (such as a plunger of a syringe) linearly by pushing the portion of the tool through the sterile drape. For some applications, a portion of the sterile drape that is disposed at the interface between the tool-actuation arm and the portion of the tool that is pushed is configured to have greater rigidity and/or wearability than other portions of the drape. For example, a sticker may be placed at the portion in order to enhance the rigidity and/or wearability of the portion relative to other portions of the sterile drape. Or, the drape may be treated (e.g., using a heat treatment, or a chemical treatment) at the portion in order to enhance the rigidity and/or wearability of the portion relative to other portions of the sterile drape. Thus, linear motion of a portion of a tool is generated by the linear tool motor, which is disposed within the non-sterile zone. The linear motion that is generated by the motor is transmitted to the portion of the tool via the drape, such as to maintain the seal between the non-sterile and sterile zones.

(14) There is therefore provided, in accordance with some applications of the present invention, apparatus for performing a procedure on a portion of a body of a patient using a tool, the apparatus including: a robotic unit including: a base; an end-effector; a tool mount configured to hold the tool; a plurality of multi-jointed arms via which the end effector is coupled to the base, each of the multi-jointed arms including a rotatable arched link in a vicinity of the end effector, the rotatable arched link being configured to accommodate rolling of the end effector about an axis that is not coaxial with a longitudinal axis of the tool.

(15) In some applications, the apparatus further includes one or more arm motors configured to move the multi-jointed arms, and a computer processor that is configured to: calculate any rolling of the end effector with respect to the base about the axis that is not coaxial with the longitudinal axis of the tool, as a result of movement of the multi-jointed arms; and drive the one or more arm motors to move the multi-jointed arms such as to compensate for rolling of the end effector about the axis that is not coaxial with the longitudinal axis of the end effector and the tool, such that the tool rolls about its own longitudinal axis.

(16) In some applications, each of the rotatable arched links defines a concavely curved surface, and the rotatable arched links are configured to accommodate rolling of the end effector by rotating such that the end effector becomes accommodated by the concavely curved surfaces of the rotatable arched links.

(17) In some applications, the apparatus further includes an end effector motor configured to directly roll the end effector with respect to the base, the rotatable arched link is configured to be rotated in a passive manner such as to accommodate the end effector being actively rolled by the end-effector motor.

(18) In some applications, the apparatus further includes a sterile drape and a drape plate, the drape plate is configured to be coupled to the end effector such that the multi-jointed arms and the end effector are disposed in a non-sterile zone on a first side of the sterile drape and the tool mount is disposed within a sterile zone on a second side of the sterile drape.

(19) In some applications, the drape plate is configured to be coupled to the end effector such that all motion-driving portions of the robotic unit that are configured to drive the end effector to move are disposed in the non-sterile zone on the first side of the sterile drape.

(20) In some applications, the rotatable arched link is configured to be rotated such as to accommodate the end effector being rolled through an angle of more than 180 degrees.

(21) In some applications, the rotatable arched link is configured to be rotated such as to accommodate the end effector being rolled through an angle of more than 300 degrees.

(22) In some applications, each of the plurality of multi-jointed arms further includes a first straight link adjacent to a first end of the rotatable arched link and a second straight link adjacent to a second end of the rotatable arched link via which the end effector is coupled to the rotatable arched link, and the second straight link is disposed at an angle with respect to the first straight link.
(23) In some applications, the apparatus further includes a motor within at least one of the arms that is configured to roll the rotatable arched link with respect to the first straight link, the angle between the first straight link and the second straight link is configured such as to cause the rolling of the rotatable arched link with respect to the straight link to result in rolling of the end effector.
(24) There is further provided, in accordance with some applications of the present invention, apparatus for performing a procedure on a portion of a body of a patient using a robotic unit that includes an end-effector and a base, a tool mount that is configured to hold a tool, a tool motor configured to roll the tool with respect to the end effector, and one or more robotic arms that are configured to move the end effector with respect to the base, the apparatus including: a drape plate configured to be placed between the tool mount and the end effector; a sterile drape disposed around and sealed with respect to the drape plate, and configured to form an interface between a non-sterile zone on a first side of the sterile drape and a sterile zone on a second side of the sterile drape, such that the tool mount is disposed within the sterile zone, and the one or more robotic arms and the tool motor are disposed within the non-sterile zone; at least one gear mechanism configured to be disposed within the sterile zone and configured to drive the tool to roll with respect to the end effector; and a motion-transmission portion configured to transmit motion from the tool motor to the at least one gear mechanism, while maintaining a seal between the sterile zone and the non-sterile zone.
(25) In some applications, the apparatus further includes at least one computer processor configured to: drive the end effector to move with respect to the base by moving the one or more arms, calculate any resultant rolling of the end effector with respect to the base, and drive the tool motor to roll the tool with respect to the end effector, such as to compensate for any resultant rolling of the end effector with respect to the base.
(26) In some applications, the motion-transmission portion includes a shaft and the tool motor is configured to drive the shaft to rotate, the at least one gear mechanism includes a first gear wheel that is driven to rotate by the shaft and a second gear wheel that is driven to rotate by the first wheel.
(27) In some applications, an interface between the shaft and the first gear wheel is sealed, such as to maintain a seal between the sterile zone and the non-sterile zone.
(28) In some applications, the first gear wheel is disposed within the drape plate.
(29) In some applications, the second gear wheel is built into the tool.
(30) In some applications, the apparatus further includes a tool sleeve configured to be disposed around the tool, the second gear wheel is built into the tool sleeve.
(31) In some applications, the motion-transmission portion includes a shaft and the tool motor is configured to drive the shaft to rotate, the at least one gear mechanism includes a worm gear that is driven to move linearly by the shaft and a gear wheel that is driven to rotate by linear movement of the first wheel.
(32) In some applications, an interface between the shaft and the worm gear is sealed, such as to maintain a seal between the sterile zone and the non-sterile zone. In some applications, the worm gear is disposed within the drape plate. In some applications, the gear wheel is built into the tool. In some applications, the apparatus further includes a tool sleeve configured to be disposed around the tool, the gear wheel is built into the tool sleeve.
(33) In some applications, the apparatus further includes: a linear tool motor configured to drive at least a portion of the tool to move linearly with respect to the end effector, a tool-actuation arm configured to be moved linearly by the linear tool motor to thereby move at least the portion of the tool linearly with respect to the end effector, the sterile drape is configured to form the interface

such that the linear tool motor is disposed within the non-sterile zone and such that tool-actuation arm is disposed within the non-sterile zone.
(34) In some applications, a portion of the sterile drape that is configured to be disposed at an interface between the tool-actuation arm and the portion of the tool that is pushed is configured to have greater rigidity and/or wearability than other portions of the drape.
(35) There is further provided, in accordance with some applications of the present invention, apparatus for performing a procedure on a portion of a body of a patient using a robotic unit that includes an end-effector, a tool mount that is configured to hold a tool such that the tool is coaxial with the end effector, a linear tool motor configured to drive at least a portion of the tool to move linearly with respect to the end effector, and one or more robotic arms that are configured to move the end effector, the apparatus including: a drape plate configured to be placed between the tool mount and the end effector; a sterile drape disposed around and sealed with respect to the drape plate, and configured to form an interface between a non-sterile zone on a first side of the sterile drape and a sterile zone on a second side of the sterile drape, such that the tool mount is disposed within the sterile zone, and the one or more robotic arms and the linear tool motor are disposed within the non-sterile zone; and a tool-actuation arm configured to be disposed within the non-sterile zone, and configured to be moved linearly by the linear tool motor to thereby move at least the portion of the tool linearly with respect to the end effector, and a portion of the sterile drape that is configured to be disposed at an interface between the tool-actuation arm and the portion of the tool that is pushed is configured to have greater rigidity and/or wearability than other portions of the drape.
(36) In some applications, the apparatus includes a sticker placed at the portion of the sterile drape the sticker being configured to enhance the rigidity and/or wearability of the portion relative to the other portions of the sterile drape.
(37) In some applications, the portion of the sterile drape is heat treated to enhance the rigidity and/or wearability of the portion relative to the other portions of the sterile drape.
(38) In some applications, the portion of the sterile drape is chemically treated to enhance the rigidity and/or wearability of the portion relative to the other portions of the sterile drape.
(39) In some applications, the portion of the sterile drape includes an alternative or additional material from the other portions of the sterile drape to enhance the rigidity and/or wearability of the portion relative to the other portions of the sterile drape.
(40) In some applications, the apparatus further includes an automatic tool-actuation arm folding mechanism that is configured to cause the tool-actuation arm to fold automatically in response to being retracted to a given distance from the tool mount.
(41) There is further provided, in accordance with some applications of the present invention, apparatus for performing a procedure on a portion of a body of a patient using a robotic unit that includes an end-effector, tool mount configured to hold a tool such that the tool is coaxial with the end effector, and a linear tool motor configured to drive at least a portion of the tool to move linearly with respect to the end effector, the apparatus including: a tool-actuation arm configured to be moved linearly by the linear tool motor, to thereby move at least the portion of the tool linearly with respect to the end effector; and an automatic tool-actuation arm folding mechanism that is configured to cause the tool-actuation arm to fold automatically in response to being retracted to a given distance from the tool mount.
(42) In some applications, the automatic tool-actuation arm folding mechanism includes a spring mechanism.
(43) In some applications, the tool includes a syringe that includes a plunger, and the tool-actuation arm is configured to push the plunger of the syringe linearly.
(44) In some applications, the tool-actuation arm is configured to fold such that the tool mount is able to accommodate a large tool without requiring removal and/or manual folding of the tool-actuation arm.

(45) In some applications, the robotic unit is configured for performing cataract surgery using a plurality of tools that include a phacoemulsification probe, and the tool-actuation arm is configured to fold such that the tool mount is able to accommodate the phacoemulsification probe without requiring removal and/or manual folding of the tool-actuation arm.
(46) In some applications, the apparatus further includes an automatic tool-actuation arm unfolding mechanism configured to cause the tool-actuation arm to automatically unfold in response to the tool-actuation arm being moved closer to the tool mount.
(47) In some applications, the automatic tool-actuation arm unfolding mechanism includes a spring mechanism.
(48) There is further provided, in accordance with some applications of the present invention, apparatus for performing a procedure on an eye of a patient using a tool, the apparatus including: a robotic unit including: a base; an end-effector; a tool mount configured to hold the tool; a plurality of multi-jointed arms via which the end effector is coupled to the base, the multi-jointed arms being configured to permit movement of the end effector with respect to the base that is such that the end effector rolls with respect to the base; at least one arm motor configured to move the multi-jointed arms; and at least one tool motor configured to rotate the tool with respect to the end effector, about a longitudinal axis of the tool; and at least one computer processor configured to: drive the arm motor to move the end effector with respect to the base by moving the multi-jointed arms, calculate any resultant rolling of the end effector with respect to the base, and drive the tool motor to roll the tool about its own longitudinal axis, such as to compensate for any resultant rolling of the end effector with respect to the base.
(49) In some applications, the robotic unit is configured to perform at least a portion of a cataract procedure on the patient's eye.
(50) The present invention will be more fully understood from the following detailed description of applications thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic illustration of a robotic system that is configured for use in a microsurgical procedure, such as intraocular surgery, in accordance with some applications of the present invention;
(2) FIGS. **2**A and **2**B are schematic illustrations of a robotic unit for use in a robotic system, in accordance with some applications of the present invention;
(3) FIGS. **3**A and **3**B are schematic illustrations of a robotic unit that is configured to roll a tool about its own axis, such as to compensate for rolling of an end effector of the robotic unit with respect to a base of the robotic unit, in accordance with some applications of the present invention;
(4) FIGS. **4**A, **4**B, and **4**C are schematic illustrations of a robotic unit having an end effector that is configured to be rolled about an axis, in accordance with some alternative applications of the present invention;
(5) FIGS. **5**A, **5**B, and **5**C are schematic illustrations of the robotic unit of FIGS. **4**A and **4**B at respective stages of a rolling motion of the end effector of the robotic unit, in accordance with some alternative applications of the present invention;
(6) FIGS. **6**A and **6**B are schematic illustrations of a robotic unit having an end effector that is configured to be rolled about an axis that is not coaxial with its own longitudinal axis, in accordance with some alternative applications of the present invention;
(7) FIGS. **7**A, **7**B, and **7**C are schematic illustrations of the robotic unit of FIGS. **6**A and **6**B at respective stages of a rolling motion of the end effector of the robotic unit, in accordance with some alternative applications of the present invention;

(8) FIG. **8** is a schematic illustration of a sterile drape and drape plate for use with a robotic unit that is not configured to rotate a tool within the end effector, in accordance with some applications of the present invention;
(9) FIG. **9** is a schematic illustration of a sterile drape and drape plate for use with a robotic unit that is configured to rotate a tool within the end effector, in accordance with some applications of the present invention;
(10) FIGS. **10**A, **10**B, and **10**C are schematic illustrations of a sterile drape and drape plate for use with a robotic unit that is configured to rotate a tool within the end effector, in accordance with some alternative applications of the present invention;
(11) FIGS. **11**A and **11**B are photographs of a sterile drape and drape plate that are generally similar to those that are schematically illustrated in FIGS. **10**A, **10**B, and **10**C, in accordance with some applications of the present invention;
(12) FIGS. **12**A and **12**B are schematic illustrations of a sterile drape and drape plate for use with a robotic unit that is configured to rotate a tool within the end effector, in accordance with some further alternative applications of the present invention;
(13) FIG. **13** is a schematic illustration of an end effector that includes an automatically foldable tool-actuation arm, for pushing a tool or a portion thereof linearly, in accordance with some applications of the present invention; and
(14) FIGS. **14**A, **14**B, and **14**C are schematic illustrations of the automatically foldable tool-actuation arm at respective stages of its motion with respect to a tool mount, in accordance with some applications of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

(15) Reference is now made to FIG. **1**, which is a schematic illustration of a robotic system **10** that is configured for use in a microsurgical procedure, such as intraocular surgery, in accordance with some applications of the present invention. Typically, when used for intraocular surgery, robotic system **10** includes first and second robotic units **20** (which are configured to hold tools **21**), in addition to an imaging system **22**, a display **24** and a control component **26** (e.g., a pair of control devices, such as joysticks, as shown), via which a user (e.g., a healthcare professional) is able to control robotic units **20**. Typically, robotic system **10** includes one or more computer processors **28**, via which components of the system and a user (e.g., a healthcare professional) operatively interact with each other. For some applications, each of first and second robotic units is supported on a base **27**, as shown. The scope of the present application includes mounting first and second robotic units in any of a variety of different positions with respect to each other.
(16) Typically, movement of the robotic units (and/or control of other aspects of the robotic system) is at least partially controlled by a user (e.g., a healthcare professional). For example, the user may receive images of the patient's eye and the robotic units, and/or tools disposed therein, via display **24**. Typically, such images are acquired by imaging system **22**. For some applications, imaging system **22** is a stereoscopic imaging device and display **24** is a stereoscopic display. Based on the received images, the user typically performs steps of the procedure. For some applications, the user provides commands to the robotic units via control component **26**. Typically, such commands include commands that control the position and/or orientation of tools that are disposed within the robotic units, and/or commands that control actions that are performed by the tools. For example, the commands may control a phacoemulsification tool (e.g., the operation mode and/or suction power of the phacoemulsification tool), and/or injector tools (e.g., which fluid (e.g., viscoelastic fluid, saline, etc.) should be injected, and/or at what flow rate). Alternatively or additionally, the user may input commands that control the imaging system (e.g., the zoom, focus, and/or x-y positioning of the imaging system). For some applications, the commands include controlling an IOL-manipulator tool, for example, such that the tool manipulates the IOL inside the eye for precise positioning of the IOL within the eye.
(17) Reference is now made to FIGS. **2**A and **2**B, which are schematic illustrations of robotic unit

20 for use in robotic system **10**, in accordance with some applications of the present invention. For some applications, each of the robotic units includes an end effector **30**. The end effector is typically configured to securely hold any one of a plurality of different tools **21** (shown in FIG. **1**) thereupon. For some applications, the end effector is coupled to a tool mount, which is configured to hold the tools (directly or indirectly), e.g., as described in further detail hereinbelow. Typically, the end effector is configured to insert the tool into the patient's eye such that entry of the tool into the patient's eye is via an incision point, and the tip of the tool is disposed within the patient's eye.
(18) For some applications, two multi-jointed arms **32** (i.e., arms containing a plurality of links **34** which are connected to each other via joints **36**) are disposed on a single side of end effector **30** and are configured to moveably support the end effector. Typically, the computer processor detects movement of the patient's eye in three dimensions, by analyzing images acquired by imaging system **22** (which as described hereinabove is typically a stereoscopic imaging system). For some applications, in response to the detected movement of the patient's eye, the computer processor drives the robotic unit to move the tool such that entry of the tool into the patient's eye remains via the incision point even as the patient's eye undergoes the movement in three dimensions. Typically, even as the patient's eye undergoes the movement in three dimensions, the computer processor drives the robotic unit to perform at least a portion of a procedure on the patient's eye by moving the tip of the tool in a desired manner with respect to the eye such as to perform the portion of the procedure, while entry of the tool into the patient's eye is maintained fixed at incision point. In this manner, the robotic unit acts to provide a dynamic remote center of motion that is located at the incision point, and about which motion of the tool is centered. Typically, the remote center of motion moves in coordination with movement of the eye. Alternatively or additionally, the computer processor is configured to detect when the eye is at a given position, and to time the performance of certain functions by the robotic units such that they are performed when the eye is at the given position.
(19) Typically, a plurality of arm-motors are associated with the two multi-jointed arms **32**. Although not shown in FIGS. **2**A and **2**B, the locations of arm motors are shown in FIGS. **3**B, **4**C, and **6**B. For some applications, the plurality of arm-motors move the end effector through five degrees-of-freedom (e.g., translational movement along the x-, y-, and z-axes, as well as pitch and yaw angular movement). It may be observed that in the example shown in FIGS. **2**A-B, each link **34** of at least one of the multi-jointed arms **32** of the robotic system includes two parallel bars **40** that extend between vertical joints **42**, with a vertical joint disposed between each pair of adjacent links. For some applications, the arrangement of parallel bars and vertical joints results in the ends of each of the joints of a given multi-jointed arm remaining parallel with each other, even as the arm moves (as shown in the transition from FIG. **2**A to FIG. **2**B). In turn, this prevents the end effector from being rolled with respect to base **27**. Put another way, the arrangement of parallel bars and vertical joints results in roll of the end effector being mechanically decoupled from translational movement and pitch-yaw angular movement of the end effector. For some applications, preventing the end effector from being rolled with respect to base **27** is desirable in order to prevent tool **21** from rolling with respect to the patient's eye. In particular, for tools that are not rotationally symmetrical it may be desirable to prevent the tools from rolling with respect to the patient's eye. For some applications, each of the multi-jointed arms is configured in the above-described manner. Alternatively, only one of the multi-jointed arms is configured in the above-described manner.
(20) Reference is now made to FIGS. **3**A and **3**B, which are schematic illustrations of robotic unit **20** that is configured to rotate tool **21** about its own axis, such as to compensate for rolling of end effector **30** of the robotic unit with respect to base **27** of the robotic unit, in accordance with some alternative applications of the present invention. (In FIG. **3**B, many of the reference numerals are not included, in order to focus specifically on the typical positions of the arm motors within the robotic unit.) As noted above, it is typically desirable to prevent a tool (and in particular a tool that

is not rotationally symmetrical) from rolling with respect to the patient's eye. In the case of procedures (such as cataract procedures) that are performed on the eye, many of the tools are not rotationally symmetrical and the dimensions of the surgical space are relatively small. For some applications, rather than preventing the end effector from rotating with respect to base **27** (e.g., as described with reference to FIGS. **2**A and **2**B), the end effector is permitted to roll with respect to the base, but such rolling of the end effector is compensated for by rolling the tool about its own axis **50** with respect to the end effector. For some applications, the robotic unit is configured to rotate a tool about its own axis, for an alternative or additional reason, e.g., in order to perform a surgical maneuver.

(21) For example, as shown in FIGS. **3**A and **3**B, neither one of multi-jointed arms **32** includes an arrangement of parallel bars, as described hereinabove with reference to FIGS. **2**A-B. Thus, the robotic unit is configured to move the end effector through six degrees-of-freedom (e.g., translational movement along the x-, y-, and z-axes, as well as pitch, yaw, and roll angular movement). Typically, the robotic unit is actively driven to move the end effector along the x-, y-, and z-axes, as well as through pitch and yaw angular movements, whereas the rolling of the end effector is an unwanted by-product of such movements. Further typically, the robotic unit includes at least five arm motors M**1**-M**5**, as shown in FIG. **3**B.

(22) For some applications, computer processor **28** (shown in FIG. **1**) calculates the amount of roll that the tool should undergo with respect to the end effector. For example, computer processor **28** may calculate that owing to movement of the multi-jointed arms (e.g., translational movement along the x-, y-, and/or z-axes, and/or pitch and/or yaw angular movement), the end effector will undergo roll of +20 degrees with respect to base. In response thereto, the computer processor may drive the tool to rotate about its own axis **50** with respect to end effector by −20 degrees. The tool is typically held by the end effector (or by a tool mount) such that the tool is coaxial with the end effector (or is coaxial with the tool mount). Therefore, longitudinal axis **50** of the tool is typically also the longitudinal axis of the end effector (or of the tool mount). Thus, in the example shown in FIGS. **3**A and **3**B, longitudinal axis **50**, around which the tool is rotated, is coaxial with the end effector longitudinal axis.

(23) As shown in FIGS. **3**A and **3**B, for some such applications, a tool motor **52** is configured to roll the tool with respect to the end effector. For some applications, the tool motor drives the tool to roll with respect to the end effector, via an arrangement of gear wheels **54**, as shown.

(24) Reference is now made to FIGS. **4**A, **4**B, and **4**C which are schematic illustrations of robotic unit **20**, end effector **30** of the robotic unit being configured to be rolled about an axis **60** that is not coaxial with its own longitudinal axis **50**, in accordance with some applications of the present invention. For some applications, the end effector is coupled with (or forms an integrated structure with) a tool mount **92**, which is configured to hold the tool, as shown in FIG. **4**B. FIG. **4**C is similar to FIG. **4**B, but many of the reference numerals are not included in FIG. **4**C, in order to focus specifically on the typical positions of the arm motors within the robotic unit. Reference is also made to FIGS. **5**A, **5**B, and **5**C, which are schematic illustrations of the robotic unit of FIGS. **4**A, **4**B and **4**C at respective stages of a rolling motion of end effector **30**, in accordance with some alternative applications of the present invention. It is noted that in some of the figures (e.g., FIGS. **5***a*-**5**C), there are symbols on robotic arms **32** (e.g., symbols A**1** and B**1**). These symbols are included in order to demonstrate the orientations of the robotic arms in respective figures.

(25) For some applications, as an alternative to or in addition to rolling the tool with respect to the end effector, the end effector itself is rolled. Typically, in such cases, the end effector is rolled about axis **60**, which is not coaxial with axis **50** of tool **21** and of end effector **30**. (Thus, the end effector undergoes eccentric rolling with respect to its longitudinal axis.) For some applications, the robotic unit includes an end-effector motor **62**, which is configured to roll the end effector about axis **60** (shown in FIG. **4**A). Typically, the robotic unit includes at least five arm motors, with the locations of the arm motors being illustrated schematically by the dashed circles labelled M**1**-M**5**, in FIG.

4C. For some applications, the computer processor drives the arms to move such as to compensate for axis **60** not being coaxial with the tool axis. In this manner, although the end effector is rotated about axis **60** via end-effector motor **62**, the tool itself is rolled about its own axis.
(26) For some such applications, each of multi-jointed arms **32** comprising a rotatable arched link **64** in a vicinity of end effector **30**. The rotatable arched link is configured to rotate such as to accommodate rolling of the end effector about axis **60**. This may be observed by observing the transition from FIG. **5**A to **5**B and then from FIG. **5**B to FIG. **5**C. As shown, as the end effector is rotated, the end effector pushes the arched link causing it to rotate, such that the end effector becomes accommodated by concavely curved surfaces **66** of the arched links. Such accommodation of the rolling of the end effector is typically desirable, particularly in view of the robotic unit being configured such that the rolling of the end effector is eccentric with respect to its own axis. For example, if in place of the rotatable arched link there was a straight link disposed perpendicularly with respect to end effector axis, then the end effector could only be rotated through a relatively small range of angle about axis **60** before being blocked by the link. By contrast, using the configuration shown in FIGS. **4**A-**5**C, the end effector is typically able to roll through more than 180 degrees, e.g., more than 250 degrees, or more than 300 degrees, about axis **60**.
(27) Reference is now made to FIGS. **6**A and **6**B, which are schematic illustrations of robotic unit **20**, end effector **30** of the robotic unit being configured to be rolled about an axis **70** that is not coaxial with its own longitudinal axis **50**, in accordance with some applications of the present invention. FIG. **6**B is similar to FIG. **6**A, but many of the reference numerals are not included in FIG. **6**B, in order to focus specifically on the typical positions of the arm motors within the robotic unit. Reference is also made to FIGS. **7**A, **7**B, and **7**C, which are schematic illustrations of the robotic unit of FIG. **6** at respective stages of a rolling motion of end effector **30**, in accordance with some alternative applications of the present invention.
(28) As described with reference to FIGS. **4**A-**5**C, for some applications, each of the multi-jointed arms includes rotatable arched link **64** in a vicinity of end effector **30**. The rotatable arched link being configured to rotate such as to accommodate rolling of the end effector about axis **70**, in a generally similar manner to that described hereinabove. For some such applications, a first straight link **80** is disposed adjacent to a first end **82** of the rotatable arched link and a second straight link **84** is disposed adjacent to a second end **86** of the rotatable arched link (with the end effector being coupled to the rotatable arched link via second straight link **84**). As shown in FIG. **6**A, for some applications, the second straight link is disposed at an angle alpha with respect to first straight link **80**.
(29) Typically, the robotic unit includes at least five arm motors, with the locations of the arm motors being illustrated schematically by the dashed circles labelled M**1**-M**5**, in FIG. **6**B. For some applications, the robotic unit includes a further motor mounted in a vicinity of straight link **80** or straight link **84** of one of the arms. For example, the robotic unit may include a further motor mounted at either the location that is indicated by the dashed circle that is labelled M**6**A or at the location that is indicated by the dashed circle that is labelled M**6**B in FIG. **6**B. The further motor is configured to roll rotatable arched link **64** with respect to straight links **80** and **84**. Typically, by virtue of the second straight links being disposed at angle alpha with respect to the first straight links, rolling of rotatable arched link **64** with respect to straight link **80** causes the second straight links to swivel with respect to the first straight links. In turn, this causes the end effector to roll about axis **70**. This may be observed in the transition from **7**A to FIG. **7**B and then from FIG. **7**B to FIG. **7**C. Typically, for such applications, computer processor **28** calculates how to move links of the arms such as to cause the end effector to roll about axis **70** in a desired manner. As described with reference to FIGS. **4**A-**5**C, as the end effector is rotated, the end effector becomes accommodated by concavely curved surfaces **66** of the arched links. Typically, using the configuration shown in FIGS. **6**A-**7**C, the end effector is able to roll (eccentrically with respect to its own axis) through more than 180 degrees, e.g., more than 250 degrees, or more than 300

degrees, about axis **70**.
(30) Reference is now made to FIG. **8**, which is a schematic illustration of a sterile drape **88** and drape plate **90** for use with a robotic unit **20** that is not configured to rotate tool **21** within end effector **30**, in accordance with some applications of the present invention. For example, a sterile drape and drape plate as shown in FIG. **8** may be used with robotic units that are as described with reference to FIGS. **4**A-C and/or with reference to FIGS. **6**A-B, whereby any rotation of the tool is typically effected by rotating the end effector, rather than rotating the tool with respect to the end effector. Typically, in such cases, all of the motion-driving portions of the robotic unit (such as motors, gear wheel, etc.) that are configured to drive the end effector to move, as well as end effector **30** itself, are disposed within a non-sterile zone on a first side of the sterile drape (i.e., on the side of the sterile drape on which the arms of the robotic unit are disposed). A tool mount **92** (which is configured to hold the tool, directly or indirectly) is disposed within a sterile zone on a second side of the sterile drape, and is couplable to the drape plate. Typically, the sterile drape is disposed around and sealed with respect to the drape plate. The drape plate is couplable to (or coupled to) both the end effector and the tool mount. For example, end effector **30**, which is disposed within the non-sterile zone at the ends of the arms, may be configured to be coupled to one side of the drape plate, and tool mount may define a portion **94** on its reverse side that is configured to become coupled to a second side of the drape plate. Typically, drape plate **90** acts as an interface between (a) arms **32** and end effector **30**, which are disposed within a non-sterile zone on a first side of the sterile drape and (b) tool mount **92** and tool **21**, which are disposed within a sterile zone on a second side of the sterile drape. When the drape plate is coupled to both the end effector and to the tool mount, movement of the arms and the end effector (which is generated within the non-sterile zone) is transmitted to tool mount **92** and to tool **21** (both of which are disposed within the sterile zone), via the drape plate. (It is noted that in some cases, the tool itself is disposed within the tool mount. Alternatively, as shown, the tool is disposed inside a tool sleeve **23**, which is disposed within tool mount **92**.)
(31) Reference is now made to FIG. **9**, which is a schematic illustration of a sterile drape **96** and drape plate **98** for use with a robotic unit in which the tool is rotated within the end effector, in accordance with some applications of the present invention. For example, a sterile drape and drape plate as shown in FIG. **9** may be used with a robotic unit as described with reference to FIGS. **3**A-B, which show an example of robotic unit **20** that is configured to rotate tool **21** about its own axis. For some such cases, at least some of the motion-driving portions of the robotic unit (such as motors, gear wheel, etc.) that are configured to drive the end effector and/or the tool to move are disposed within the sterile zone (i.e., the side of the drape that is shown in FIG. **9**). For some applications, tool motor **52** and/or gear wheels **54**A and **54**B (which are configured to roll the tool with respect to the end effector) are disposed within the sterile zone. (It is noted that in some cases, the tool itself includes an in-built gear wheel, and is directly rotated by gear wheel MA that is driven to rotate by the motor. Alternatively, as shown, the tool is disposed inside tool sleeve **23** that includes or is coupled to gear wheel MB, which is rotated by gear wheel MA.) For some applications, a linear tool motor **100** that is configured to drive a portion of the tool to move linearly is disposed within the sterile zone. Linear tool motor is typically configured to move a portion of a tool (such as a plunger **120** of a syringe) linearly via a tool actuation arm **110**. Some examples of the linear tool motor and tool actuation arm are described in further detail hereinbelow.
(32) Typically, all portions of the apparatus that are configured to be disposed within the sterile zone are configured to be disposable and/or sterilizable (e.g., via autoclaving). For applications as shown in FIG. **9**, typically tool motor **52**, gear wheel **54**A, tool sleeve **23** (and gear wheel **54**B), linear tool motor **100**, and tool actuation arm **110** are all configured to be disposable and/or sterilizable (typically, via autoclaving). Typically, tool motor **52** and linear tool motor **100** are powered via a sealed electrical connector that passes through the sterile drape and/or by an external cable.

(33) Typically, sterile drape **96** is disposed around and sealed with respect to drape plate **98**. For some applications, arms **32** and end effector **30** (arms and end effector not shown in FIG. **9**) are disposed within the non-sterile zone, and drape plate **98** is couplable to the end effector. Typically drape plate **98** acts as an interface between (a) end effector **30** and arms **32** (which are not shown in FIG. **9** and) which are disposed within a non-sterile zone on a first side of the sterile drape, and (b) tool mount **92** and tool **21** which are disposed within a sterile zone on a second side of the sterile drape. When the drape plate is coupled to both the end effector and to the tool mount, then movement of the arms and the end effector (which is generated within the non-sterile zone) is transmitted to tool mount **92** and to tool **21** (both of which are disposed within the sterile zone), via drape plate **98**. However, as described above, for applications such as that shown in FIG. **9**, movement of the tool (or a portion thereof) with respect to the end effector is effected from within the sterile zone, via tool motor **52** and/or linear tool motor **100**.
(34) Reference is now made to FIGS. **10**A, **10**B, and **10**C, which are schematic illustrations of a sterile drape **102** and drape plate **104** for use with a robotic unit **20** that is configured to rotate tool **21** within end effector **30**, in accordance with some alternative applications of the present invention. Typically drape plate **104** acts as an interface between (a) arms **32** (which are not shown in FIGS. **10**A-C and) and end effector **30**, which are disposed within a non-sterile zone on a first side of the sterile drape and (b) tool mount **92** and tool **21** which are disposed within a sterile zone on a second side of the sterile drape.
(35) For some applications, tool motor **52** (shown in FIGS. **10**B-C) is disposed on end effector **30**, within the non-sterile zone. Tool motor **52** typically directly drives a motion-transmission portion **106** (such as a pin or a shaft) to rotate. The motion-transmission portion is configured to transmit rotational motion of the motor to first gear wheel (i.e., spur gear) **54**A and the first gear wheel drives the tool to rotate with respect to the end effector by driving second gear wheel (i.e., spur gear) **54**B to rotate. For some applications, the first gear wheel is disposed within (e.g., built into) the drape plate. As described hereinabove, second gear wheel **54**B can be built into the tool itself or can be built into or coupled to tool sleeve **23**. Typically, motion-transmission portion **106** is mechanically coupled to first gear wheel **54**A in such a manner that the interface between the motion-transmission portion and first gear wheel **54**A is sealed such as to maintain a seal between the sterile zone and the non-sterile zone (e.g., via an O-ring **108**, as shown in FIG. **10**C). Thus, in the example shown in FIGS. **10**A-C, rotational motion of the tool with respect to the end effector is generated by motor **52**, which is disposed within the non-sterile zone. The rotational motion that is generated by the motor is transmitted to the tool via an interface that maintains the seal between the non-sterile and sterile zones.
(36) Referring to FIG. **10**A, for some applications, linear tool motor **100** is disposed within the non-sterile zone. Linear tool motor **100** typically drives tool-actuation arm **110** to move linearly. For such applications, tool-actuation arm **110** is typically disposed within the non-sterile zone and is configured to push a portion of a tool (such as a plunger **120** of a syringe) linearly by pushing the portion of the tool through sterile drape **102**. For some applications, a portion **114** of the sterile drape that is disposed at the interface between the tool-actuation arm and the portion of the tool that is pushed is configured to have greater rigidity and/or wearability than other portions of the drape. For example, a sticker **116** may be placed at portion **114** in order to enhance the rigidity and/or wearability of the portion relative to other portions of the sterile drape. Or, the drape may be treated (e.g., using a heat treatment, or a chemical treatment) at portion **114** in order to enhance the rigidity and/or wearability of the portion relative to other portions of the sterile drape. Or, the drape may comprise an alternative or additional material at portion **114** as compared to other portions of the drape in order to enhance the rigidity and/or wearability of the portion relative to other portions of the sterile drape. Thus, in the example shown in FIGS. **10**A-C, linear motion of a portion of a tool is generated by linear tool motor **100**, which is disposed within the non-sterile zone. The linear motion that is generated by the motor is transmitted to the portion of the tool via the drape, such as

to maintain the seal between the non-sterile and sterile zones.
(37) Typically, sterile drape **102** is disposed around and sealed with respect to drape plate **104**. Typically, drape plate **104** is couplable to the end effector and is coupled to (or couplable to) tool mount **92**. When the drape plate is coupled to both the end effector and to the tool mount, then movement of the arms and the end effector (which is generated within the non-sterile zone) is transmitted to the tool mount and to the tool (both of which are disposed within the sterile zone), via the drape plate.
(38) Reference is now made to FIGS. **11**A and **11**B, which are photographs of a sterile drape **102** and drape plate **104** that are similar to those that are schematically illustrated in FIGS. **10**A, **10**B, and **10**C, in accordance with some applications of the present invention. FIG. **11**A is a photograph showing the view of the sterile drape and drape plate from the sterile zone. As may be observed, tool mount **92** is shown, tool mount **92** being built into drape plate **104** in the example that is shown. In addition, sticker **116** is shown. As described above, the sticker is configured to be placed at a portion of the sterile drape that is disposed at the interface between the tool-actuation arm and the portion of the tool that is pushed is configured to have greater rigidity and/or wearability than other portions of the drape. It may also be observed that drape **102** is shaped such as to be placed over arms of a robotic unit. FIG. **11**B is a photograph showing the view of the sterile drape and drape plate from the non-sterile zone. As may be observed, the back side of the drape plate is typically shaped to define a housing **122**. The housing typically houses gear wheel **54**A. Housing portion is typically configured to be coupled to end effector **30** (shown in FIG. **10**A), which is disposed at the end of the arms and supports tool motor **52**. For example, the housing portion may be coupled to the end effector via a snap-lock mechanism.
(39) Reference is now made to FIGS. **12**A and **12**B, which are schematic illustrations of a sterile drape **124** and drape plate **126** for use with a robotic unit in which the tool is rotated within the end effector, in accordance with some further alternative applications of the present invention. The apparatus as shown in FIGS. **12**A-B is generally similar to that shown and described with respect to FIGS. **10**A-C, except for the following differences. In the apparatus shown in FIGS. **12**A-B, tool motor **52** is configured to drive a worm gear **128** to move linearly (e.g., in an up-down direction), in order to drive gear wheel **54**B (which is typically built into or coupled to tool **21** or tool sleeve **23**) to rotate. As described with reference to FIGS. **10**A-C, typically, tool motor **52** is disposed upon end effector **30**, which is disposed within the non-sterile zone. Tool motor **52** typically directly drives a linear motion-transmission portion **130** (such as a pin or a shaft) to move linearly (e.g., in an up-down direction). The motion-transmission portion is configured to transmit linear motion of the motor to worm gear **128** and the worm gear drives the tool to rotate with respect to the end effector by driving gear wheel (i.e., spur gear) **54**B to rotate. For some applications, the worm gear is disposed within (e.g., built into) the drape plate. Typically, linear motion-transmission portion **130** is mechanically coupled to worm gear **128** in such a manner that the interface between the linear motion-transmission portion and worm gear **128** is sealed such as to maintain a seal between the sterile zone and the non-sterile zone (e.g., via an O-ring **132**, as shown in FIG. **12**B). Thus, in the example shown in FIGS. **12**A-B, motion of the tool with respect to the end effector is generated by motor **52**, which is disposed within the non-sterile zone. Linear motion that is generated by the motor is transmitted to the sterile zone via an interface that maintains the seal between the non-sterile and sterile zones. The linear motion is then converted to rotational motion of the tool with respect to the end effector.
(40) Reference is now made to FIG. **13**, which is a schematic illustration of end effector **30** that includes tool-actuation arm **110** for pushing a tool or a portion thereof linearly, in accordance with some applications of the present invention. For some applications, the tool-actuation arm is configured to fold automatically in response to being retracted to a given distance from tool mount **92**. Reference is also made to FIGS. **14**A, **14**B, and **14**C, which are schematic illustrations of an automatically foldable tool-actuation arm at respective stages of its motion with respect to a tool

holding portion of the end effector, in accordance with some applications of the present invention. As described hereinabove, typically, tool-actuation arm **110** is configured to push a portion of a tool (such as plunger **120** of a syringe) linearly. Typically, linear tool motor **100** drives the arm to move linearly via a transmission shaft **134** (shown in FIG. **13**). For some applications, the tool-actuation arm is configured to fold automatically in response to being retracted to a given distance from tool mount **92**, as shown in the transition from FIG. **14**A to FIG. **14**B and then from FIG. **14**B to FIG. **14**C. In this manner, the tool-actuation arm may be folded automatically such as to accommodate the insertion of a larger tool, such as a phacoemulsification probe, into the tool mount, without requiring removal and/or manual folding of the tool-actuation arm. Typically, the tool-actuation arm is configured to fold automatically by means of an automatic tool-actuation arm unfolding mechanism, such as a spring mechanism, being activated. Further typically, in response to the tool-actuation arm being moved closer to the tool mount, the tool-actuation arm is configured to automatically unfold (e.g., via an automatic tool-actuation arm unfolding mechanism, such as a spring mechanism, being activated). For some applications, rather than being configured to fold automatically, the arm is configured to be moved in a different manner such as to accommodate the insertion of a larger tool, such as a phacoemulsification probe, into the tool mount, without requiring removal and/or manual movement of the tool-actuation arm. For example, the arm may be configured to automatically retracted, e.g., using an electromechanical actuator, a spring mechanism, etc.

(41) It is noted that the scope of the present applications includes combining elements of the sterile drape, the drape plate, and the tool actuation arm that are shown in respective figures with each other. Purely by way of example, the tool actuation arm shown in FIGS. **13-14**C can be combined with any one of the examples of sterile drapes and drape plates described with reference to FIGS. **9-12**B.

(42) Although some applications of the present invention are described with reference to cataract surgery, the scope of the present application includes applying the apparatus and methods described herein to other medical procedures, mutatis mutandis. In particular, the apparatus and methods described herein to other medical procedures may be applied to other microsurgical procedures, such as general surgery, orthopedic surgery, gynecological surgery, otolaryngology, neurosurgery, oral and maxillofacial surgery, plastic surgery, podiatric surgery, vascular surgery, and/or pediatric surgery that is performed using microsurgical techniques. For some such applications, the imaging system includes one or more microscopic imaging units.

(43) It is noted that the scope of the present application includes applying the apparatus and methods described herein to intraocular procedures, other than cataract surgery, mutatis mutandis. Such procedures may include collagen crosslinking, endothelial keratoplasty (e.g., DSEK, DMEK, and/or PDEK), DSO (descemets stripping without transplantation), laser assisted keratoplasty, keratoplasty, LASIK/PRK, SMILE, pterygium, ocular surface cancer treatment, secondary IOL placement (sutured, transconjunctival, etc.), iris repair, IOL reposition, IOL exchange, superficial keratectomy, Minimally Invasive Glaucoma Surgery (MIGS), limbal stem cell transplantation, astigmatic keratotomy, Limbal Relaxing Incisions (LRI), amniotic membrane transplantation (AMT), glaucoma surgery (e.g., trabs, tubes, minimally invasive glaucoma surgery), automated lamellar keratoplasty (ALK), anterior vitrectomy, and/or pars plana anterior vitrectomy.

(44) Applications of the invention described herein can take the form of a computer program product accessible from a computer-usable or computer-readable medium (e.g., a non-transitory computer-readable medium) providing program code for use by or in connection with a computer or any instruction execution system, such as computer processor **28**. For the purpose of this description, a computer-usable or computer readable medium can be any apparatus that can comprise, store, communicate, propagate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device. The medium can be an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system (or apparatus or device) or a

propagation medium. Typically, the computer-usable or computer readable medium is a non-transitory computer-usable or computer readable medium.
(45) Examples of a computer-readable medium include a semiconductor or solid-state memory, magnetic tape, a removable computer diskette, a random-access memory (RAM), a read-only memory (ROM), a rigid magnetic disk and an optical disk. Current examples of optical disks include compact disk-read only memory (CD-ROM), compact disk-read/write (CD-R/W), DVD, and a USB drive.
(46) A data processing system suitable for storing and/or executing program code will include at least one processor (e.g., computer processor **28**) coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code in order to reduce the number of times code must be retrieved from bulk storage during execution. The system can read the inventive instructions on the program storage devices and follow these instructions to execute the methodology of the embodiments of the invention.
(47) Network adapters may be coupled to the processor to enable the processor to become coupled to other processors or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.
(48) Computer program code for carrying out operations of the present invention may be written in any combination of one or more programming languages, including an object-oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the C programming language or similar programming languages.
(49) It will be understood that the algorithms described herein, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer (e.g., computer processor **28**) or other programmable data processing apparatus, create means for implementing the functions/acts specified in the algorithms described in the present application. These computer program instructions may also be stored in a computer-readable medium (e.g., a non-transitory computer-readable medium) that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instruction means which implement the function/act specified in the algorithms. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the algorithms described in the present application.
(50) Computer processor **28** is typically a hardware device programmed with computer program instructions to produce a special purpose computer. For example, when programmed to perform the algorithms described with reference to the Figures, computer processor **28** typically acts as a special purpose robotic-system computer processor. Typically, the operations described herein that are performed by computer processor **28** transform the physical state of a memory, which is a real physical article, to have a different magnetic polarity, electrical charge, or the like depending on the technology of the memory that is used. For some applications, operations that are described as being performed by a computer processor are performed by a plurality of computer processors in combination with each other.
(51) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present

invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. Apparatus for performing a procedure on a portion of a body of a patient using a robotic unit that includes an end-effector, tool mount configured to hold a tool such that the tool is coaxial with the end effector, and a linear tool motor configured to drive at least a portion of the tool to move linearly with respect to the end effector, the apparatus comprising: a tool-actuation arm configured to be moved linearly by the linear tool motor, to thereby move at least the portion of the tool linearly with respect to the end effector; and an automatic tool-actuation arm folding mechanism that is configured to cause the tool-actuation arm to fold automatically in response to being retracted to a given distance from the tool mount.
2. The apparatus according to claim 1, wherein the automatic tool-actuation arm folding mechanism comprises a spring mechanism.
3. The apparatus according to claim 1, wherein the tool includes a syringe that includes a plunger, and wherein the tool-actuation arm is configured to push the plunger of the syringe linearly.
4. The apparatus according to claim 1, wherein the tool-actuation arm is configured to fold such that the tool mount is able to accommodate a large tool without requiring removal and/or manual folding of the tool-actuation arm.
5. The apparatus according to claim 1, wherein the robotic unit is configured for performing cataract surgery using a plurality of tools that include a phacoemulsification probe, and wherein the tool-actuation arm is configured to fold such that the tool mount is able to accommodate the phacoemulsification probe without requiring removal or manual folding of the tool-actuation arm.
6. The apparatus according to claim 1, further comprising an automatic tool-actuation arm unfolding mechanism configured to cause the tool-actuation arm to automatically unfold in response to the tool-actuation arm being moved closer to the tool mount.
7. The apparatus according to claim 6, wherein the automatic tool-actuation arm unfolding mechanism comprises a spring mechanism.
8. A method for performing a procedure on a portion of a body of a patient using a robotic unit that includes an end-effector, a tool mount that is configured to hold a tool such that the tool is coaxial with the end effector, and a linear tool motor configured to drive at least a portion of the tool to move linearly with respect to the end effector, the method comprising: driving a tool-actuation arm to move linearly, to thereby move at least the portion of the tool linearly with respect to the end effector; and causing the tool-actuation arm to fold automatically by driving the tool-actuation arm to become retracted to a given distance from the tool mount, such as to actuate an automatic tool-actuation arm folding mechanism.
9. The method according to claim 8, wherein the automatic tool-actuation arm folding mechanism comprises a spring mechanism.
10. The method according to claim 8, wherein the tool includes a syringe that includes a plunger, and wherein driving the tool-actuation arm to move linearly comprises pushing the plunger of the syringe linearly.
11. The method according to claim 8, wherein causing the tool-actuation arm to fold comprises accommodating a large tool within the tool mount without requiring removal or manual folding of the tool-actuation arm.
12. The method according to claim 8, wherein causing the tool-actuation arm to fold comprises accommodating a phacoemulsification probe within the tool mount without requiring removal or manual folding of the tool-actuation arm.
13. The method according to claim 8, further comprising actuating an automatic tool-actuation arm

unfolding mechanism configured to cause the tool-actuation arm to automatically unfold by driving the tool-actuation arm to move closer to the tool mount.

14. The method according to claim 13, wherein the automatic tool-actuation arm unfolding mechanism comprises a spring mechanism.